

Worksheet 4

Math 1A

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Problem Q1

Suppose f is a function and L and a are real numbers.

未自动求解: 概念题需要 LLM 推理 (可接入 Qwen/Kimi API)

- (a) 概念题需要 LLM 推理 (可接入 Qwen/Kimi API)
- (b) 概念题需要 LLM 推理 (可接入 Qwen/Kimi API)
- (c) 概念题需要 LLM 推理 (可接入 Qwen/Kimi API)
- (d) 概念题需要 LLM 推理 (可接入 Qwen/Kimi API)

Problem Q2

Is ∞ a number? What does $\lim_{x \rightarrow a} f(x) = \infty$ mean? Does $\lim_{x \rightarrow a} f(x) = \infty$ mean that the limit exists?

Solution.

∞ 不是实数。

$\lim_{x \rightarrow a} f(x) = \infty$ 表示 $f(x)$ 在 $x \rightarrow a$ 时无界增长 (发散), 这个极限在通常意义下不存在。

Final Answer:

∞

Problem Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

Solution.

夹逼定理 (Squeeze Theorem): 若 $g(x) \leq f(x) \leq h(x)$ 在 a 附近成立, 且 $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, 则 $\lim_{x \rightarrow a} f(x) = L$ 。

几何直观: f 被上下两个函数“夹住”, 两者都收敛到 L , f 也必收敛到 L 。

Final Answer:

L

Problem Q4

What do we mean by $\lim_{x \rightarrow a^-} f(x) = L$? What is meant by $\lim_{x \rightarrow a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Find an algebraic expression for such a function.

Solution.

$$\lim_{x \rightarrow a} f^2 x^2 = a^2 f^2$$

Final Answer:

$$a^2 f^2$$

Problem P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exist.

Solution.

(无法直接计算: 'list' object has no attribute 'has')

$$\lim_{x \rightarrow a} f x = a f$$

$$\lim_{x \rightarrow a} g x = a g$$

Final Answer:

$$a g$$

Problem P2

(a) Find $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x \rightarrow 0^-} (\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{|x|})$, if they exist.

Solution.

$$\lim_{x \rightarrow 0} -\frac{1}{|x|} + \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Final Answer:

$$0$$

Problem P3

For what kinds of functions can one evaluate the limit $\lim_{x \rightarrow a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \rightarrow a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

未自动求解: 物理题: 无法识别具体公式, 请补充已知量

Problem AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

Solution.

Final Answer:

见子题

$$(a) \lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7x^5 - 24}} = -20 \quad -20$$

$$(b) \lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10 \quad 10$$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ 0

Problem AP2

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

Solution.

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$$

Final Answer:

π

Problem AP3

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

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